

Central nervous system pathology in patients with the Guillain–Barré syndrome

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Summary

Thirteen autopsy cases of patients with clinical criteria of the Guillain–Barré syndrome were investigated for pathological changes and cellular composition of inflammatory infiltrates in the CNS and PNS. The survival times from the onset of neurological symptoms until death ranged from 1 day to 12 months. In the CNS, degeneration of spinal posterior tracts was seen in three cases. Mononuclear infiltrates consisted of evenly proportioned lymphocytes and macrophages in cases with survival of 1 and 2 days, whereas macrophages predominated in cases with survival of 5 days and longer. Infiltrates presented as nodular clusters around blood vessels and neurons, or were scattered diffusely. They were found within the spinal cord in eight out of 13 cases, within the medulla oblongata in eight out of 12 cases, within the pons in five out of nine cases, and in one out of four midbrains. Activation of microglia, either focal or diffuse, was found in various degrees in 11 out of 13 cases, involving the spinal cord (six out of 13 cases), the medulla oblongata (10 out of 12 cases), the pons (five out of nine cases) or as subependymal rims along the walls of the ventricular system and the central canal of the spinal cord (seven out of 13 cases). In the PNS, myelin loss (12 out of 13 cases), axonal degeneration (six out of 13 cases) and mononuclear cell infiltrates (13 out of 13 cases) were seen in segmental and cranial nerves, spinal

ganglia and spinal roots in varying distribution and severity. Mononuclear cell infiltrates were composed of macrophages and T lymphocytes, with even distribution in cases with short survival (1 and 2 days), and predominance of macrophages in cases with protracted clinical course. T lymphocytes were equally composed of OPD4+ and CD8+ cells without obvious differences between cases of short and long duration, or between PNS and CNS infiltrates in 11 out of 12 cases, whilst two cases had a dominant OPD4+ subset. We conclude that CNS pathology is frequent in patients with Guillain–Barré syndrome. It involves axons with secondary myelin impairment, microglial activation and inflammatory infiltration. In this series, primary demyelination is not found in the CNS. Changes such as degeneration of spinal posterior tracts are secondary to pathology in the PNS. Inflammatory cell reactions in the CNS are similar to those in the PNS and to CNS pathology in experimental allergic neuritis. This inflammation might reflect CNS immune activation in the absence of the relevant antigen, in addition to cellular reactions accompanying secondary CNS changes. The presence of distinct pathology in the CNS is in contrast with other recent studies on the pathology of Guillain–Barré syndrome which, unlike this study, may have been influenced by recently introduced treatments.

Keywords: Guillain–Barré syndrome; central nervous system; cell infiltrates; lymphocyte subsets; neuropathology

Abbreviations: cDNA = complementary DNA; CMV = cytomegalovirus; EAN = experimental allergic neuritis; HSV = herpes simplex virus; VZV = varicella zoster virus

Introduction

Neuropathological reports on CNS pathology in patients with Guillain–Barré syndrome are rare and are mostly based on a small number of cases. In recent studies, CNS

changes were interpreted as secondary to the pathological process in the PNS (peripheral nervous system), and were found to be restricted to central chromatolysis of spinal

anterior horn cells and degeneration of posterior columns (Dehaene *et al.*, 1986; Feasby *et al.*, 1986; Kanda *et al.*, 1989; Honavar *et al.*, 1991). Only in earlier studies (Haymaker and Kernohan, 1949; Asbury *et al.*, 1969; van Zandycke *et al.*, 1982) were some sparse and non-characteristic findings of primary CNS pathology reported, including perivascular lymphocytes in the white matter and subependymal parenchyma of the brain, proliferation of meningeal cells or inflammatory infiltration of meninges. However, histopathological analysis was then restricted to conventional staining techniques; no information was given about the amount of pathological material available in each case, and poor clinical information raised some doubt on the accuracy of the diagnosis (Hughes, 1990).

In this retrospective histopathological study we examined the CNS in a series of Guillain-Barré syndrome cases, all untreated by intravenous immunoglobulins, and most untreated by corticosteroids.

Material and methods

Post-mortem material

Thirteen autopsy cases of patients were investigated at the Institute of Neurology, Vienna, between 1957 and 1985 (Cases 2–4, 7–11, 13), and at the Neuropathological Laboratory of the Institute of Pathology, Innsbruck, between 1987 and 1991 (Cases 1, 5, 6, 12). Metabolic, toxic, neoplastic or paraneoplastic syndromes as differential diagnosis to Guillain-Barré syndrome were excluded *intra vitam* and/or post-mortem by complete autopsy in all cases. The clinical records were reviewed for correctness of the diagnosis; all cases fulfilled the criteria for Guillain-Barré syndrome (Asbury *et al.*, 1978). Clinical data of all cases are given in Table 1.

Since this study was retrospective, the sampling of tissues was not standardized. Spinal cord and spinal roots were examined in all 13 cases; if available, samples from forebrain (11 cases), oblongata (12 cases), pons (nine cases),

Table 1 Clinical data of 13 patients with Guillain-Barré syndrome

Case no. (death)	Duration of signs and symptoms	Age	Sex	Steroid therapy	Plasma- pheresis	Antecedent event	Tetraparesis/ -plegia	Cranial nerve involvement	Notes	Cause of death
1 (1991)	1 day	78	F	No	No	None	Lower limb paresis	?	Cardiac failure	Pulmonary embolism
2 (1974)	2 days	67	M	No	No	Operation for tongue carcinoma	Yes	No		Respiratory arrest
3 (1970)	5 days	59	F	No	No	None	Yes	No		Pulmonary embolism
4 (1985)	6 days	41	M	Yes	No	Upper respiratory tract infection	Yes	Yes (bulbar, III, IV, VI, VII)		Pneumonia
5 (1990)	9 days	78	M	Yes	No	Influenza-like illness	Yes	Yes (III, IV, VI)		Sepsis
6 (1990)	14 days	56	F	No	Yes	Cough	Yes	Yes (III, IX, X, XI, XII)	Pneumothorax	Acute respiratory failure
7 (1970)	3 weeks	71	M	No	No	Arterial bypass	Yes	?	Acute respiratory failure	Endocarditis
8 (1980)	4 weeks	79	M	No	No	Influenza-like illness	Yes	Yes (bulbar)	Cardioversion, respiratory failure	Cardiorespiratory failure
9 (1978)	5 weeks	65	M	No	No	Influenza-like illness	Yes	Yes	Respiratory failure	Respiratory arrest
10 (1958)	6 weeks	26	M	No	No	None	Yes	Yes (bulbar, VII)		Respiratory arrest
11 (1971)	8 weeks	19	M	No	No	Epistaxis, fever	Yes	Yes (bulbar)		Pneumonia
12 (1987)	12 months	77	M	No	No	None	Yes	No		Respiratory failure
13 (1957)	?	16	M	No	No	None	Yes	No		Asystoly

Table 2 Antibodies used for immunocytochemistry

Antibody	Clone	Antigen recognised	Labelled target	Dilution	Source
Anti-LCA	PD7/26, 2B11	CD45RB antigen, leukocyte common antigen	Leukocytes	1:10	DAKO A/S, Glostrup, Denmark
Anti-CD3	Polyclonal	CD3 antigen	T cells	1:50 (pp)	DAKO
Anti-CD8	C8/144B	CD8 antigen	Cytotoxic/suppressor T cells	1:40	DAKO
Anti-CD20	L26	CD20 antigen	B cells	1:50 (ap)	DAKO
OPD4	OPD4	CD45RO antigen	Activated T (and B) cells	1:20 (pp)	DAKO
Anti-CD45RO	UCHL-1	CD45RO antigen	Activated and (in part) resting T cells, granulocytes, monocytes	1:150	DAKO
Anti-CD68	KP1	CD68 antigen	Macrophages	1:100(pp)	DAKO
Anti-Ki-M1P	Ki-M1p	Ki-M1P antigen	Monocytes/macrophages, microglia (Gehrmann and Kreutzberg, 1993)	1:10 (pp)	Dr Wacker, Kiel, Germany (Radzun <i>et al.</i> , 1991)
Anti-MBP	Polyclonal	MBP	Myelin	1:100	DAKO
Anti-NFP	2F11	NFP, mainly phosphorylated	Neuronal cells + processes	1:100	DAKO
Anti-GFAP	Polyclonal	GFAP	Normal and reactive astrocytes	1:400	DAKO
Anti-HSV-1	Polyclonal	HSV type 1 antigens	HSV type 1	1:100	DAKO
Anti-HSV-2	Polyclonal	HSV type 2 antigens	HSV type 2	1:100	DAKO
Anti-CMV	CCh2	CMV antigens	CMV	1:20	DAKO
Anti-VZV GP 01	8-1	Glycoprotein 3 of VZV	VZV	1:2000	Dr Kurata, Tokyo, Japan
Anti-VZVCI/170.1	17-1	Nucleoprotein of VZV	VZV	1:1000	Dr Kurata, Tokyo, Japan
Anti-measles virus	Polyclonal	Measles virus antigens	Measles virus	1:100	Dr Vainionpää, Turku, Finland
KV Hochsterwitz	Polyclonal	arboviral antigens	TBME virus	1:20	Dr F. X. Heinz, Vienna, Austria

LCA = leukocyte common antigen; pp = protease pretreatment; ap = autoclave pretreatment at 120°C and 1 bar for 10 min in citrate buffer pH 6.0; MBP = myelin basic protein; NFP = neurofilament proteins; GFAP = glial fibrillary acidic protein; HSV = herpes simplex virus; CMV = cytomegalovirus; VZV = varicella zoster virus; TBME = central European tick-borne meningoencephalitis.

mesencephalon (four cases), peripheral nerves (six cases), cranial nerves (11 cases), spinal ganglia (four cases), and skeletal muscle (three cases) were investigated. The type of peripheral nerves and exact segment levels of spinal ganglia could not be identified in all cases. Cervical levels of the spinal cord were available in 12 out of 13 cases, lumbar levels in 11 cases, thoracic levels in 10 cases, sacral segments and cauda equina in each of seven cases.

Formalin-fixed, paraffin-embedded sections were stained with haematoxylin and eosin, Klüver-Barrera's method for myelin, Bodian silver impregnation for axons and by immunocytochemical methods.

Immunocytochemistry

Table 2 summarizes the antibodies used. Immunostaining of sections was performed using the peroxidase-anti-peroxidase method with polyclonal antibodies, and the labelled biotin-avidin method with monoclonal antibodies.

Antiviral immunocytochemistry and in situ hybridization

All cases were immunohistochemically investigated for presence of viral antigens (*see* Table 2). *In situ* hybridization for herpes simplex virus (HSV) types I and II, cytomegalovirus (CMV) and varicella zoster virus (VZV) was performed

on up to three representative tissue blocks in all cases. Biotinylated complementary DNA (cDNA) probes for HSV I and II, and CMV (Enzo Biochem, New York, USA) and DNA probes for VZV (VZGP 01, VZVCI/170.1, obtained from Dr Kurata, Tokyo) were used and processed as described in previous studies (Schmidbauer *et al.*, 1989a, b, 1992). All anti-viral antibodies and DNA probes have been working reliably on paraffin sections of positive control tissues.

Semiquantitative grading of cellular infiltrates and of tissue damage

Evaluation of the severity of cellular infiltrates and of demyelination or axonal pathology was performed by grading inflammatory infiltrates as absent (0); very scarce (0/+); mild (+); moderate (++) and severe (+++). Scarce (0/+) infiltrates were not considered to be pathological changes as they were similar in intensity to control cases (*see* below).

For assessment of lymphocyte subtypes, approximately even amounts of OPD4+ and CD8+ cells were valued as + in both subsets. In cases with differing composition, the dominating subset was estimated as ++, the minor one as +.

The severity of demyelination and of axonal damage was estimated as absent (0); mild (+) = <25% of fibres affected; moderate (++) = 26–50% of fibres affected and severe (+++) = more than 50% of fibres affected.

Table 3 Cell infiltrates in peripheral and central nervous system and changes in skeletal muscle

Case no. (year)	Duration of illness	T cells (CD3, UCHL-1)	OPD4+ cells	CD8+ cells	B cells (CD20)	Macrophages (LCA, Ki-M1P)	Microglia in CNS (LCA, Ki-M1P)	Muscle
1 (1991)	1 day	+ SR, SC ++ SG	PNS: ly. +	PNS: ly. +	+ (scarce)	+ SR, ++ SG + SC (perivasc.)	0	—
2 (1974)	2 days	+ SC (perivascu.) ++ MO, + pons, thalamus +/++ SR, PN, SM	CNS: ly. ++ PNS: ly. +	CNS: ly. + PNS: ly. +	+ (scarce in PN, MO)	+ thalamus, MO, pons + SC (perivasc.) +/++ PN, SM	Periventr. (+ temporal CX, BG ++ pons, MO, SC) ++ spinal grey	—
3 (1970)	5 days	+ men., SR	CNS: ly. perivascular + PNS: ly. +	CNS: ly. perivascular + PNS: ly. +	+ (scarce in men.)	+ MO, +/++++ SC, SR, +/++ CN, +++ PN, + men.	+/++ MO (n. gracilis, periventr.), SC (periventr.)	—
4 (1985)	6 days	+ MO, BG, SC, +/++ SR, + SG, SM	CNS: ly. + PNS: ly. +	CNS: ly. + PNS: ly. +	+ (scarce)	+ MO, SC, SR +/++++ SG	+ MO (nodular, diffuse), ++ SC (periventr., lat. col., grey matter)	—
5 (1990)	9 days	+/++ cauda, PN, SM	CNS: ly. + PNS: ly. +	CNS: ly. (+) PNS: ly. +	0	+/++++ SR, cauda, SM, PN	+ pons, MO (periventr., olives)	—
6 (1990)	14 days	+ MO, +/++ SR, SM, +++ SG, ++ PN	CNS: ly. ++ PNS: ly. +	CNS: ly. + PNS: ly. + men.: ly. +	0	+/++ SR, anterior horn, men., +++ SG, ++ PN	+++ MO (periventr., focal), + SC	—
7 (1970)	3 weeks	+ SR	CNS: ly. + PNS: ly. +	CNS: ly. +, PNS: ly. +	0	+/++++ SRs + MO (perivasc.)	+ MO (nodules, periventr.)	—
8 (1980)	4 weeks	+ SR, SG, SM	CNS: ly. + PNS: ly. +	CNS: ly. PNS: ly. +	+ (scarce)	+ CN, SG, SM +/++++ SR	+ mesencephalon	Rare regenerating fibres, no infiltrates
9 (1978)	5 weeks	0/+ SR	CNS: ly. + PNS: ly. +	CNS: ly. + PNS: ly. +	0	+ MO, SR +/++ CN	+ MO (nodules)	Groups of atrophic fibres, myonecroses Focally mph.
10 (1958)	6 weeks	+ SR	—	—	0	+/++++ SR	0/+ SC (periventr.)	—
11 (1971)	8 weeks	+/++ CN ++ SR + PN	PNS: ly. +	PNS: ly. +	0	+/++++ CN +++ SR, PN ++ sacral dura	+/++ SC (perivasc., diffuse)	Groups of atrophic fibres. Fibrosis. Perivasc. ly., mph. In endo+ perimysium
12 (1987)	12 mths	0 CN + SR	CNS: mph. +		0	0/+ CN +/++ SC, +/++++ SR	+ SC (lat. col., grey matter)	—
13 (1957)	?	+ brain stem, SR, PN	CNS: ly. + PNS: ly. +	CNS: ly. + PNS: ly. +	0	+/++ brainstem, CN, SR, PN	+/++ pons, MO, SC	—

SR = spinal roots; SC = spinal cord; SG = spinal ganglion; SM = spinal meninges; MO = medulla oblongata; PN = peripheral nerve; CN = cranial nerves; CX = cortex; BG = basal ganglia; perivasc. = perivascular; periventr. = periventricular; lat. col. = lateral column; men. = meninges; ly. = lymphocytes; mph. = macrophages. Changes graded as follows: 0 = none; 0/+ = very scarce; + = slight; ++ = moderate; +++ = severe. — = not examined.

Controls

Five control cases were used for evaluation of incidence of infiltrating mononuclear cells and of the density of microglia and as negative controls for immunocytochemistry. We investigated sections from brain and spinal cord with roots of patients who died immediately (or <3 h) after a fatal event (diagnoses: acute brainstem haemorrhage due to hypertension; suicide by shooting; myocardial infarction; subarachnoid bleeding following rupture of an aneurysm; and acute ischaemia due to anaesthesia accident).

Results

Composition of cellular infiltrates

Mononuclear cell infiltrates were similar in PNS and CNS, and were composed of lymphocytes and macrophages in varying density, distribution and proportion (Table 3). Ten out of 13 cases showed predominance of monocytes/macrophages. In cases with duration of 1 and 2 days, T lymphocytes and macrophages were present in almost even densities, as was also found in one case with a survival of

Table 4 *Changes in the spinal cord*

Case no. (duration)	Spinal cord										
	Central chromatolysis of motor neurons	Posterior tract degeneration	Spinocerebellar tract degeneration	Mononuclear cell infiltrates				Microglial activation			
				Posterior tracts	Lateral column	Grey matter	Meninges	Posterior tracts	Lateral column	Grey matter	Pericanalicular
1 (1 day)	0	0	0	+	0	+	0	0	0	0	0
2 (2 days)	+	+	0	+	+	+	++	0	+	++ (AH)	++
3 (5 days)	0	0	0	0/+	0	0	0/+	0	0	+	++
4 (6 days)	+	0	0	0	0/+	+	+	0	+	+/++	++
5 (9 days)	0	0	0	0	0	0	+/++	0	0	0	0
6 (14 days)	0	0	0	+	0/+	+	+/++	0	0	0/+	0
7 (3 weeks)	++	0	0	0	0	0	0	0	0	0	0
8 (4 weeks)	0	0	0	0/+	0	0/+	0/+	0	0	0/+	0
9 (5 weeks)	+	0	0	+	+	0	0	0	0	0/+	0
10 (6 weeks)	0	0	0	0	0	0	0	0	0	0	0/+
11 (8 weeks)	++	+	0	+/++	0/+	++	0	+	+	++	0/+
12 (12 mths)	0	++	0	++	0/+	+	0	0	+	+	0
13 (?)	+	0	0	0	0	+	0	0	0	+	0

AH = anterior horn; D = diffuse. Changes graded as follows: 0 = none; 0/+ = very scarce; + = slight; ++ = moderate.

Table 5 *Changes in the brainstem*

Case no. (duration)	Medulla oblongata			Pons			Mesencephalon		
	Microglial activation	Mononuclear infiltrates	ML degen.	Microglial activation	Mononuclear infiltrates	ML degen.	Microglial activation	Mononuclear infiltrates	ML degen.
1 (1 day)	0	0	0	—	—	—	—	—	—
2 (2 days)	++ dorsolateral diffuse and periventr.	+ perivascular	0	++ peri-ventricular + diffuse	+ perivascular	0	—	—	—
3 (5 days)	+ nodules ++ n. gracilis, ++ periventr.	0/+ diffuse, meninges	0	+ periventr.	0	0	—	—	—
4 (6 days)	+ diffuse, nodular, periventr.	++ perivascular	0	0/+ scarce	+ perivascular	0	—	—	—
5 (9 days)	+ nodules, periventr.	+ nodular in olives	0	+ periventr.	+ perivascular	0	—	—	—
6 (14 days)	+ periventr.	+ perivascular	0	0	0	0	—	—	—
7 (3 weeks)	+ nodules, periventr.	0/+ scarce perivascular	0	0	0/+ scarce perivascular	0	—	—	—
8 (4 weeks)	+ periventr.	0	0	—	—	—	0	+ perivascular, scattered	0
9 (5 weeks)	+ nodules	+ perivascular	0	—	—	—	0	0	0
10 (6 weeks)	—	—	—	—	—	—	—	—	—
11 (8 weeks)	+++ diffuse	+ perivascular, ++ n. gracilis, olive	0	++ focal	+ focal, scattered	0	0	0	0
12 (12 mths)	0/+	+ nodular in olives	0	0	0	0	0	0	0
13 (?)	+ diffuse	++ perivascular	0	+ diffuse	+/++ perivascular, scattered	0	—	—	—

ML = medial lemniscus; degen. = degeneration; n. = nucleus; periventr. = periventricular. Changes graded as: 0 = none; 0/+ very scarce; + slight; ++ moderate, +++ severe; — = not examined.

Table 6 *Changes in the forebrain*

Case no. (duration)	Chiasma opticum			Basal ganglia		Temporal cortex, hippocampus	
	Demyelination	Axonal damage	Infiltrates	Microglia activation	Mononuclear cell infiltrates	Microglia activation	Mononuclear cell infiltrates
1 (1 day)		—		0	0	0	0
2 (2 days)		—		+ periventr. + thalamus	+ thalamus (perivascular)	+ periventr.	0
3 (5 days)		—		0	0	0	0
4 (6 days)		—		0	+ perivascular	0	0
5 (9 days)		—		0	0	0	0
6 (14 days)		—		0	0	0	0
7 (3 weeks)		—		0	0	0	0
8 (4 weeks)	0	0	0	0	0	0	0
9 (5 weeks)	0	0	0	0	0	0	0
10 (6 weeks)		—		—	—	—	—
11 (8 weeks)		—		0	0	0	0
12 (12 mths)		—		0	0	0	0
13 (?)		—		—	—	—	—

Periventr. = periventricular; 0 = no changes; + = slight changes; — = not examined.

14 days. B lymphocytes were either not present (eight cases) or rare (five cases).

Changes in the CNS

Type, distribution and extent of changes in the CNS of all cases are summarized in Tables 4–6.

Mononuclear cell infiltrates

Infiltrates consisted of macrophages and lymphocytes in varying intensity, localization and infiltration patterns. Macrophages were detected among motoneurons of the anterior horn (Fig. 1A) or among neurons of the nucleus olivaris inferior (Fig. 1B), in nodular clusters (Fig. 2), perivascularly (Fig. 3), or diffusely scattered.

Three Guillain-Barré syndrome cases with short-term survival showed perivascular cuffing of lymphocytes within the spinal cord or medulla oblongata (Fig. 4A–F).

Immunostaining for T cell subtypes using antibodies against OPD4+ and CD8+ cells showed additional immunostaining of microglial cells in the CNS, and rarely of macrophages (one case), with anti-OPD4. Lymphocytic infiltrates consisted of both OPD4+ and CD8+ cells in even proportions, with the exception of two cases with survival times of 2 days and 2 weeks, which showed a distinctly stronger OPD4+ component.

In 11 cases, the composition of lymphocytic infiltrates with OPD4+ and CD8+ cells was similar in the PNS and CNS; only in Case 6, did the intensity of CNS infiltration by OPD4+ cells seem to exceed that by CD8+ cells, whilst in the PNS lymphocytic subsets were evenly distributed.

Microglial activation

In 11 out of 13 cases, activation of microglia was prominent in the CNS. Microglial distribution was either diffuse or

nodular, the topographical distribution and density varied between cases.

Microglial activation was found predominantly in the grey matter of the spinal cord (Fig. 1A), and in some cases in dorsolateral quadrants or in the nuclei of the posterior tracts of the medulla oblongata (Fig. 2), within nuclei of the pontine grey (Fig. 5) and in the walls of the fourth ventricle (Fig. 6A) and/or surrounding the central canal of the spinal cord (Fig. 6B).

Secondary changes due to PNS damage

CNS damage secondary to PNS pathology was indicated by axonal reaction/central chromatolysis of motoneurons in the anterior horn of the spinal cord and in three cases by secondary degeneration of the posterior columns.

Changes in the PNS

Table 7 gives an overview of cell infiltrates, degree of demyelination and axonal damage in various sites of the peripheral nervous system of all cases.

It was not possible to detect any obvious relationship between intensity and/or distribution of cell infiltrates and the duration or severeness of illness.

Immunohistochemistry and in situ hybridization for viral infections

No specific reaction products were detected.

Control cases

In spinal roots, no (or very scarce) mononuclear cells were found within the fascicles. Few CLA- and/or CD68-positive

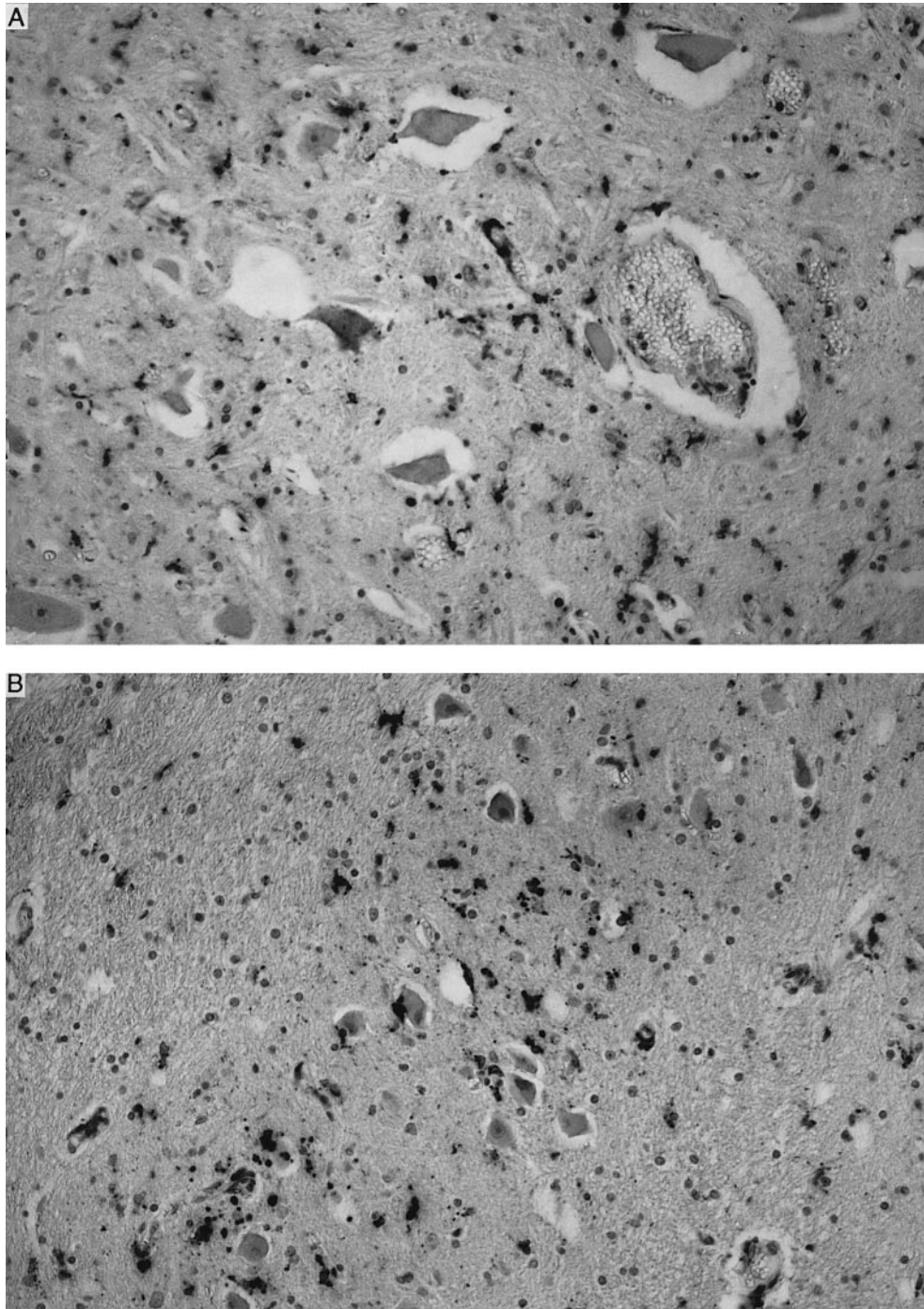


Fig. 1 Case 11. (A) Anterior horn of the spinal cord with infiltration by microglial cells and macrophages. Anti-CD68, magnification $\times 200$. (B) Macrophage clusters among neurons of the nucleus olivaris inferior of medulla oblongata. Anti-CD68, magnification $\times 200$.

cells were seen around vessels in the parenchyma of the spinal cord and forebrain. Microglial cells were seen in low density within the CNS, but two cases showed a slightly higher density (+) of microglial cells in the wall of the fourth ventricle.

Discussion

In this study on Guillain-Barré syndrome pathology, changes within the CNS were seen frequently in varying degrees of intensity and distribution of infiltrating macrophages, microglial cells, and/or lymphocytes. Lymphocytic infiltrates

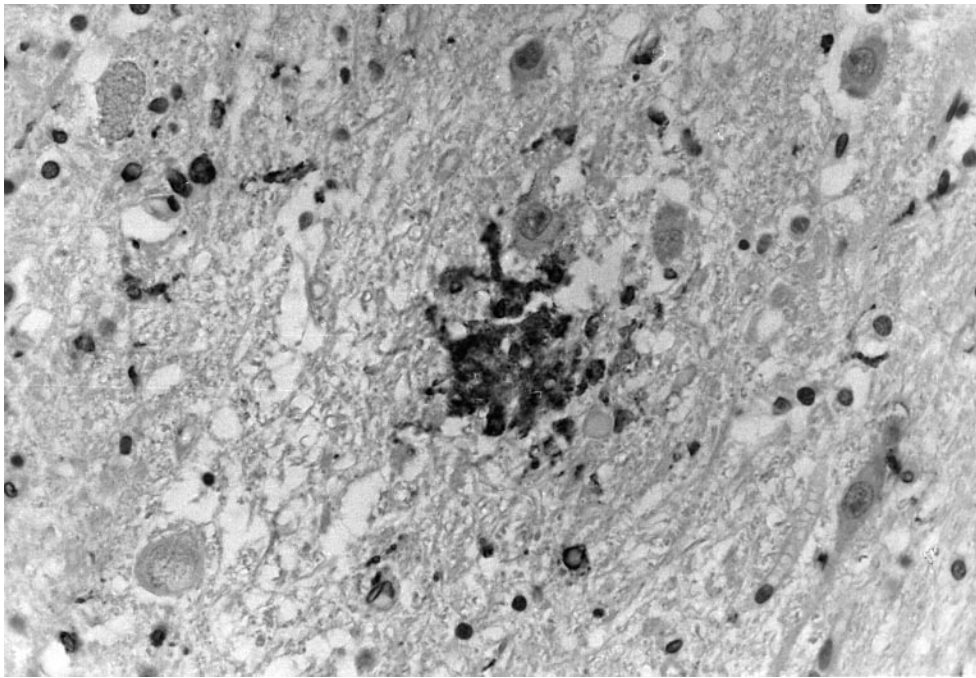


Fig. 2 Case 5. Nodular cluster of microglial cells in the dorsolateral medulla oblongata (nucleus gracilis). Anti-Ki-M1P, magnification $\times 400$.

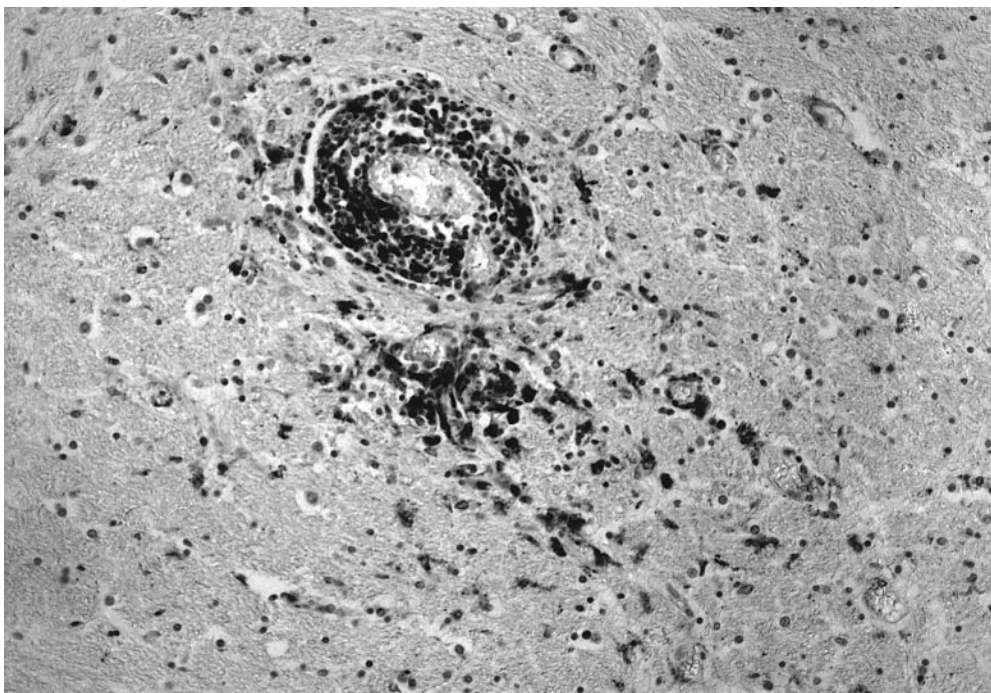


Fig. 3 Case 4. Perivascular and nodular aggregates of macrophages. Anti-CD68; magnification $\times 100$.

were found within the spinal cord or brainstem in cases with short survival, up to 6 days. The presence of T lymphocytes in the CNS was neglected in some studies on experimental allergic neuritis (EAN) (Izumo *et al.*, 1985), but was observed in another study in which increased numbers of macrophages were present from the onset in the dorsal columns of the spinal cord, accompanied by axonal loss, myelin degeneration and reactive gliosis with significant increase in the number

of T lymphocytes and Ia-positive cells in the CNS (Lassmann *et al.*, 1991).

In normal brain tissue, perivascular microglial cells are a subpopulation of resident intraparenchymal microglia (Lassmann *et al.*, 1991). We found activation of microglia in the CNS predominantly in the spinal cord and brainstem, in one case as early as 2 days after onset of neurological symptoms. In two cases, microglial activation was seen in

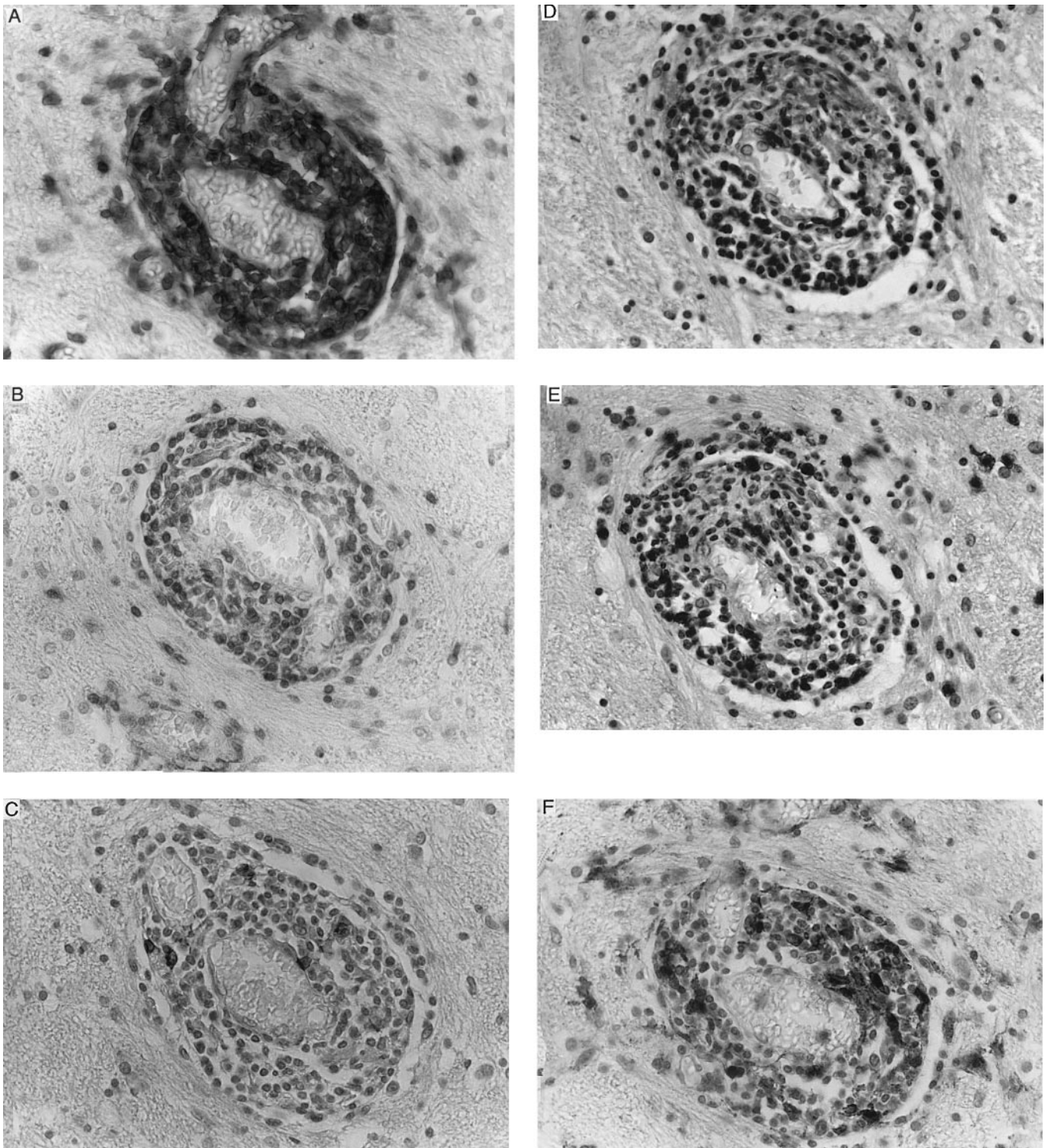


Fig. 4 Serial sections of the same vessel in medulla oblongata of Case 4. (A) Perivascular cuffing of mononuclear cells reactive with anti-LCA. (B) T-cell component, anti-CD3. (C) Scarce B lymphocytes, anti-CD20. (D) OPD4-labelled cells. (E) CD8+ lymphocytes. (F) Intermingled macrophages. Anti-CD68; magnifications $\times 264$.

the nuclei of the posterior tracts and around neurons of the spinal anterior horn. Infiltrating microglia has been described as one major feature in the spinal cord of EAN rats with a peak at ~ 7 days after T cell transfer (Gehrmann *et al.*, 1992);

microglial cells were scattered throughout the ventral and dorsal grey matter of the spinal cord, or clustered around motor neurons, and were described as appearing in increased number in the projection fields of primary afferent sensory

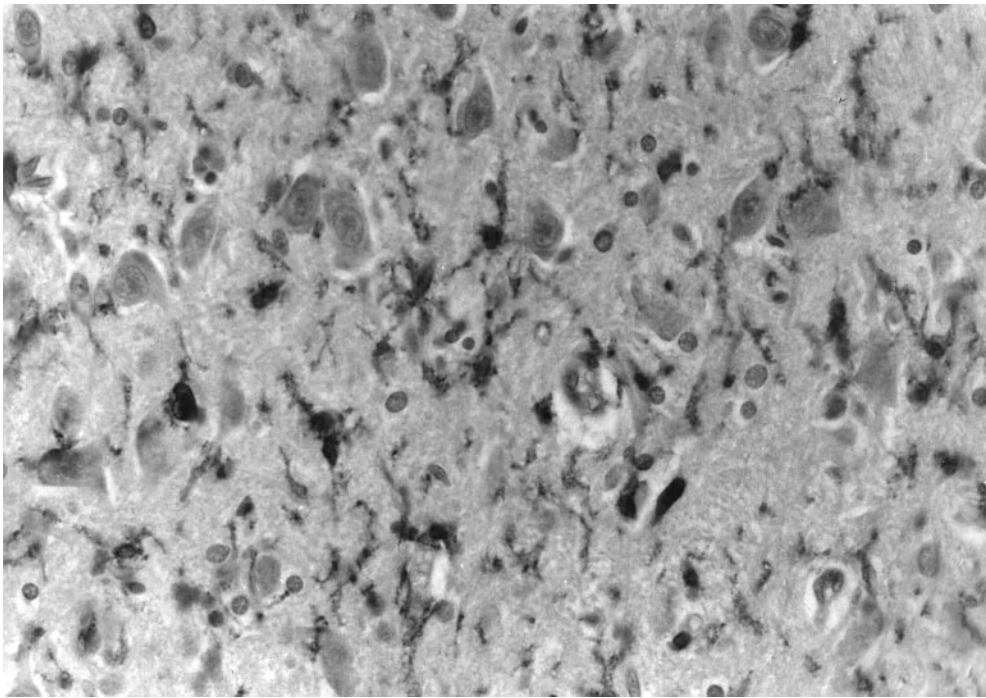


Fig. 5 Microglial activation in pontine nuclei of Case 11. Anti-LCA, magnification $\times 400$.

fibres of the lumbar (nucleus gracilis) and cervical (nucleus cuneatus) spinal segments of the rat in perineuronal and perivascular positions (Gehrmann *et al.*, 1992). Thus, inflammatory features seen in our Guillain–Barré syndrome series closely resemble those of its experimental model. However, periventricular microglial activation as found in this series has not been described in EAN. We found it in three out of four cases with short survival time (up to 6 days). In cases of longer duration, periventricular microglial infiltrates were absent or only mild, similar in intensity to those found in some of our control cases. It might be speculated that antigen presentation in the CSF of Guillain–Barré syndrome cases at an early stage might provoke severe periventricular microglial activation, but that it decreases with increasing duration of illness.

It is possible that microglia activation in the CNS, as described in this series, reflects a non-specific reaction, similar to that seen in patients after long lasting intensive care unit therapy (Schmidbauer *et al.*, 1989a). However, microglial cells were already present after only a few days in nearly all our short survival cases.

The Miller–Fisher syndrome is a variant of Guillain–Barré syndrome with characteristic clinical symptoms; several studies have suggested a brainstem lesion as the morphological substrate of clinical signs and symptoms (Al-Din *et al.*, 1982; Barontini and Sitá, 1983; Ropper, 1983; Phillips *et al.*, 1984; Dehaene *et al.*, 1986). However, only one clinicopathological study described a case as exhibiting ‘areas of central necrosis, occasionally rod cells, and astrocytic proliferation, sometimes forming ‘glial stars’, and perivascular cuffing of mononuclear cells’ (Al-Din *et al.*, 1982). However, other authors have doubted the existence

of such a specific brainstem lesion; the pathological process does not necessarily have to be localized in the brainstem to cause clinical features of the Miller–Fisher syndrome (Barontini and Sitá, 1983; Ropper, 1983; Phillips *et al.*, 1984). In our series, additional clinical involvement of cranial nerves was similar to the Miller–Fisher syndrome in three cases (Cases 4–6), but it was not possible to gain more detailed clinical information on additional cerebellar dysfunction retrospectively. We found pathological changes in the brainstem of these cases, featuring microglia activation and/or macrophage infiltration.

In some clinical studies on patients with chronic inflammatory demyelinating polyneuropathy using CT or MRI, CNS lesions were interpreted as foci of demyelination (Rubin *et al.*, 1987; Thomas *et al.*, 1987; Ormerod *et al.*, 1990) but their true nature remains obscure. There are only a few neuropathological studies reporting the combination of demyelination in both the PNS and CNS (Lassmann *et al.*, 1981; Best, 1985; Tachi *et al.*, 1985; de la Monte *et al.*, 1986). Because of its rarity, the combination of primary myelin damage in both CNS and PNS is an unusual event, but it may indicate possible links between the pathogenesis of Guillain–Barré syndrome and multiple sclerosis. In other words, both entities might represent endpoints of a clinicopathological continuum of CNS and PNS disease (Poser, 1987; Thomas, 1992). In this study, however, central demyelination was absent in all cases.

Pathology of the PNS in patients with Guillain–Barré syndrome has been well defined (van Zandycke *et al.*, 1982; Phillips *et al.*, 1984; Best, 1985; Dehaene *et al.*, 1986; Kanda *et al.*, 1989; Honavar *et al.*, 1991; for review, see Hughes, 1990). The composition of cellular infiltrates and

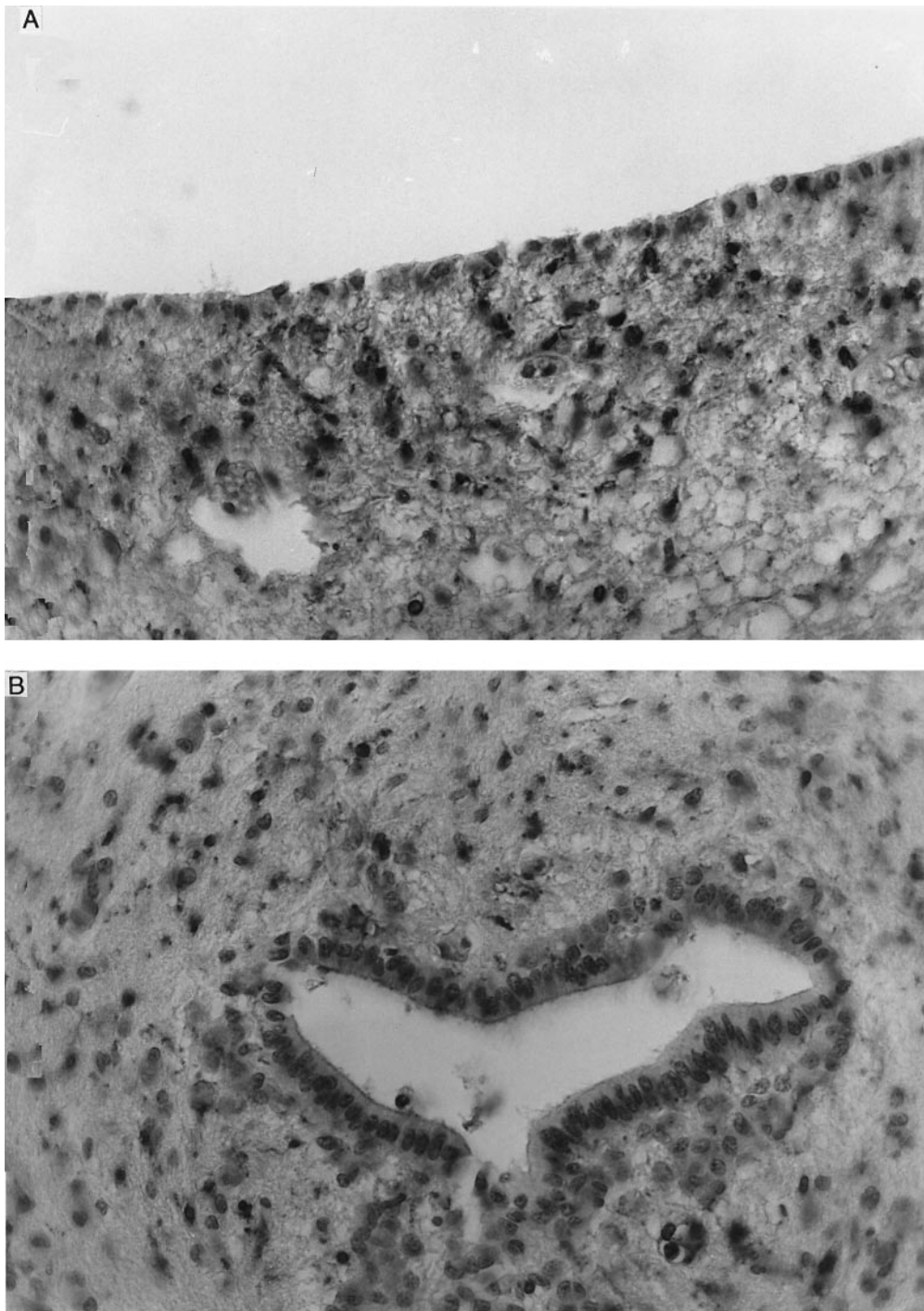


Fig. 6 Microglial activation, Case 3. (A) Floor of the 4th ventricle in medulla oblongata. (B) Central canal of cervical spinal cord. Anti-LCA; magnifications $\times 400$.

their distribution in cases with very short clinical course (of <5 days) have so far not been analysed. This study confirms that macrophages dominate the inflammatory PNS changes, accompanied by a more or less conspicuous component of T lymphocytes. Even after clinical courses as short as 1 or 2 days, T cells and macrophages are present in nearly even densities. An earlier study, including cases with short survival of a few days, described lymphocytic infiltrates in the PNS

of all cases (Asbury *et al.*, 1969). Other, more recent studies, however, report a more variable presence of lymphocytes in Guillain-Barré syndrome lesions (Carpenter, 1972; van Zandycke *et al.*, 1982; Honavar *et al.*, 1991) including cases with predominant macrophage infiltrates, but absent lymphocytes (Kanda *et al.*, 1989). In biopsy studies of Guillain-Barré syndrome, lymphocytic infiltration was not a frequent observation (Prineas *et al.*, 1972; Brechenmacher

Table 7 Neuropathological changes in the peripheral nervous system: (A) mononuclear cell infiltration, (B) myelin loss and (C) axonal damage

Case no. (duration)	Mononuclear cell infiltration										SG	PN	CN
	Anterior root				Posterior root								
	C	T	L	S	C	T	L	S	CE				
(A) Mononuclear cell infiltration													
1 (1 day)	+	—	++	—	+	—	+	—	+	++	—	IX/X 0	
2 (2 days)	0	0/+	0	0	0/+	0/+	+	+	—	—	+/++	V +	
3 (5 days)	+	++	+++	+++	++	++	+++	+++	+	—	+++/>+++	IX/X +, XII ++	
4 (6 days)	0	0	+	0	+	+	++	+	++	+++	—	IX/X 0	
5 (9 days)	0	—	0	—	0	—	+/++	—	++	—	+++	XII 0	
6 (14 days)	0	—	0	—	+	—	++	—	++	+++	+++	—	
7 (3 weeks)	0	+	+	+	+	+++	++	++	—	—	—	0	
8 (4 weeks)	0	0	—	+	0	0/+	—	++	+++	++	—	III +	
9 (5 weeks)	0	0	+	0	0	0	0	0	0	—	—	IX/X ++	
10 (6 weeks)	—	+++	+++	—	—	+++	+++	—	—	—	—	—	
11 (8 weeks)	0/+	0	+	+	++	++	++	+++	—	—	+++	III +, V +++, IX/X +	
12 (12 mths)	+	++	0	—	+++	+++	++	—	—	—	—	IX/X -/+	
13 (?)	+	++	—	—	++	++	—	—	—	—	0	V +, XII ++	
(B) Myelin loss													
1 (1 day)	0	—	0	—	0	—	0	—	+	+	—	IX/X 0	
2 (2 days)	0	0	0	0	0	0	0/+	0/+	—	—	+/++	V 0	
3 (5 days)	0	0	0	0	0/+	0	0	0	0	—	+	IX/X 0, XII 0	
4 (6 days)	0	0	++	0	0	+	++	+	++	+++	—	IX/X 0	
5 (9 days)	0	—	0	—	0	—	0	—	0/+	—	++	XII 0	
6 (14 days)	0	—	0	—	0	—	+	—	++	+	+++	—	
7 (3 weeks)	0/+	0/+	++	++	+	+	+++	++	—	—	—	0	
8 (4 weeks)	0	0	—	0	0	0	—	+	++	0/+	—	III 0	
9 (5 weeks)	0	0	0	0	0	0	0	0	0	—	—	0	
10 (6 weeks)	—	+/++	+/++	—	—	+/++	+/++	—	—	—	—	—	
11 (8 weeks)	0	0	+	+	++	+	++	+++	—	—	+++	III ++, V ++, IX/X +/++	
12 (12 mths)	0	+	0	—	0	+	0/+	—	—	—	—	IX/X +	
13 (?)	0	0	—	—	++	++	—	—	—	—	0	0	
(C) Axonal damage													
1 (1 day)	0	—	0	—	0	—	0	—	0	0	—	IX/X 0	
2 (2 days)	0	0	0	0	0	0	0	0	—	—	+	V 0	
3 (5 days)	0	0	0	0	0	0	0	0	0	—	+	IX/X 0, XII: 0	
4 (6 days)	0	0	0	0	0	0	0	0	0	0	—	IX/X 0	
5 (9 days)	0	—	0	—	0	—	0	—	0/+	—	+	XII 0	
6 (14 days)	0	—	0	—	0	—	0	—	0/+	0	0	—	
7 (3 weeks)	0	0	+	+	0	+	+	+	—	—	—	0	
8 (4 weeks)	0	0	—	0	0	0	—	0/+	+	0	—	III 0	
9 (5 weeks)	0	0	0	0	0	0	0	0	0	—	—	0	
10 (6 weeks)	—	0	0	—	—	0	0	—	—	—	—	—	
11 (8 weeks)	0	0	+	+	0	0	++	++	—	—	++	III ++, V 0, IX/X +	
12 (12 mths)	0	0	0	—	0	0	0	—	—	—	—	IX/X 0	
13 (?)	0	0	—	—	0	0	—	—	—	—	0	0	

SG = spinal ganglia; PN = peripheral nerve; CN = cranial nerve. C = cervical, T = thoracic, L = lumbar, S = sacral, CE = cauda equina. Neuropathology severity: 0 = none; 0/+ very scarce; + slight; ++ moderate; +++ severe; — = not examined.

et al., 1987). This reported variation in observations might reflect some sampling bias, but more likely indicates that Guillain-Barré syndrome pathology is heterogeneous with topographically varying severity of changes (Hughes, 1990). This is also supported by the considerable variation seen in our own series.

Our study presents new data on T-cell subsets in a larger

series of Guillain-Barré syndrome cases, taking advantage of the antibodies applicable on formalin-fixed, paraffin-embedded material, which have not been available until recently. Frozen sections could be investigated in one case of an autopsy study, describing the presence of both CD4+ and CD8+ cells and a probable dominating CD4+ subset in some nerves (Griffin *et al.*, 1993). Nerve biopsies have been

examined in two cases, showing a stronger CD4+ component and a less intense CD8+ fraction in a Guillain-Barré syndrome patient, and dominating CD8+ cells in a patient with HIV polyneuropathy (Cornblath *et al.*, 1987). In EAN, the presence of CD4+ and CD8+ cells has been reported in the PNS, but not within the spinal cord (Gehrmann *et al.*, 1992).

CD4 labelling data need to be interpreted carefully in formol-fixed tissue: the monoclonal antibody OPD4 which works on paraffin sections was originally described as labelling CD4+ helper/inducer T cells (Yoshino *et al.*, 1989). More recently, this same antibody was shown to recognize mainly the CD45RO antigen (Poppema *et al.*, 1991). In our cases, both OPD4+ and CD8+ lymphocytes occur among infiltrating cells, mostly evenly distributed, in both PNS and CNS, also in cases with very short duration of illness. In multiple sclerosis, acute plaques show predominant infiltration by CD4+ cells, and chronic active lesions are associated with a dominating CD8+ subset (Raine, 1994). In our Guillain-Barré syndrome series, only one case with survival time of 14 days showed a dominating OPD4+ cell component in the CNS, but evenly distributed T-cell subsets in the PNS.

In our series, we did not observe an obvious primary axonal lesion as reported in rare cases of acute (Feasby *et al.*, 1986; Feasby *et al.*, 1993) and chronic idiopathic polyneuropathy (Julien *et al.*, 1989). In the skeletal muscle, neurogenic damage featuring grouped atrophy of muscle fibres (Phillips *et al.*, 1984), as also seen in two out of three cases of this series, is evidence of axonal impairment, which is likely to be secondary.

We conclude that the CNS is frequently affected in our patients with Guillain-Barré syndrome. The CNS changes are non-demyelinating and therefore do not represent an equivalent of the PNS myelin damage in Guillain-Barré syndrome. Changes such as degeneration of spinal posterior tracts or perineuronal microgliosis in projection nuclei are secondary to pathology in the PNS. Inflammatory cell reactions are also found in the CNS and are similar to those in the PNS. Moreover, they are similar to CNS pathology in EAN of the rat, including infiltrates of mononuclear cells and activation of microglia. This inflammation might reflect CNS immune activation in the absence of the relevant antigen, in addition to cellular reactions accompanying secondary CNS changes. These changes might relate to clinical signs of CNS involvement in patients with Guillain-Barré syndrome.

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